

GenAI Guardrail Factory

A CI/CD-style safety release gate that blocks unsafe enterprise GenAI deployments before production — by adversarially testing, scoring, auto-remediating, and re-validating AI outputs.

Presented by **Sana Iqbal** · GenAI Solution Architect, AI Services (BFSI)
TCS^AI Google Hackathon 2026 · Track: Build with Vertex AI (Google ADK)

DESCRIPTION

Enterprises are shipping GenAI assistants to production faster than they can validate them. Today that validation is manual — sample prompts, expert review, ad-hoc checks — which doesn't scale and leaves no audit trail. **Guardrail Factory wraps any target GenAI app as a pre-production release gate.** A Google ADK evaluator agent generates adversarial tests, scores each response for groundedness, toxicity, and PII risk against the app's own source documents, and a deterministic gate decides **block or approve**. On failure it diagnoses the cause, hardens the prompt, and re-tests — approving only after measurable recovery. The verdict is governed: the same inputs always produce the same decision, with full evidence stored for audit.

KEY USE CASES

- **Pre-release safety gating** — automated block/approve verdict for any GenAI app before it ships.
- **Regression safety testing** — re-validate after every prompt, model, or document change.
- **Auto-remediation** — diagnose failures and harden prompts, then re-test to recovery.
- **Governance & audit evidence** — store scores, thresholds, verdicts, and timestamps per run.
- **Reusable safety accelerator** — one gate across BFSI assistants, compliance copilots, and RAG workflows.

TARGET PERSONAS

- **AI deployment teams** — a pass/fail gate in the release pipeline.
- **Compliance & risk leaders** — repeatable, evidence-backed sign-off.
- **CISOs & security teams** — PII, jailbreak, and policy-boundary defense.
- **Business / product owners** — confidence that a release is safe to ship.

BUSINESS IMPACT

Replaces a 6–18 month manual safety effort with an automated, repeatable gate; cuts red-team toil; and turns “it worked when we tested it” into auditable proof. Documented proof-of-concept run:

8 → 0 active failures cleared (PoC run)	0.84 → 0.94 groundedness after hardening	0.99 PII protection in final run	0 PII leaks in final run
---	--	--	------------------------------------

Outcome: a release that should have been stopped **was stopped**, then recovered to a measurable, evidence-backed pass — reducing breach and compliance exposure before production (avg. cost of a data breach in India, IBM 2025: ₹22Cr).

GOOGLE PRODUCTS & TECHNOLOGIES USED

Google ADK (LlmAgent) orchestrates the agentic evaluation · **Gemini 2.5 Flash** powers test generation, failure diagnosis, and prompt hardening · **Vertex AI** evaluation patterns · **Cloud Run** hosts the deployed service · **BigQuery + Looker** provide the production-scale governance path. Supporting: FastAPI orchestrator, ChromaDB vector store for RAG grounding, SQLite RunStore for PoC run history.

Model Armor protects the live road. Guardrail Factory decides whether the vehicle is safe enough to enter it — complementary controls across the GenAI lifecycle.